HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and engineered **Python** data pipelines on **AWS Lambda** and **S3** to process approximately **3,500 events/sec** during peak hours, replacing a less efficient scheduled-batch job.
- Developed and deployed **Java Spring Boot** REST APIs for secure inter-service data exchange, incorporating request validation and circuit-breaker patterns to enhance system resilience.
- Containerized microservices using **Docker** and orchestrated deployments to a **Kubernetes** cluster (EKS), managing Helm chart configurations and resource allocation for predictable rollouts.
- Translated ambiguous technical challenges into clearly defined problems and targeted experiments, evaluating different systems and architectural approaches through prototyping and benchmarking.
- Designed scalable, reliable infrastructure to support both traditional **ML** and emerging **AI** workloads, assessing architectural trade-offs across performance, scalability, and efficiency.
- Collaborated with engineering and research teams to understand platform requirements and identify shared infrastructure needs, ensuring alignment with broader organizational goals.
- Implemented monitoring and alerting using **Splunk** and **Dynatrace** for enterprise applications, managing SLAs and responding to P1/P2 incidents to ensure operational stability.

SKILLS

Languages – Python, Java, SQL, R, Bash

Frameworks & Libraries – Spring Boot, scikit-learn, pandas, NumPy, FastAPI, Flask

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

ML & AI – Machine Learning, Model Training, Model Evaluation, Experiment Design, Distributed Systems, ML Infrastructure

Tools & Platforms – Splunk, Dynatrace, Jira, Confluence

BI & Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling

PROJECTS

ML Infrastructure Design for Experimentation Platform

- Architected and built a shared **ML infrastructure** on **AWS** to facilitate rapid experimentation for research teams, integrating **Python** pipelines for data ingestion and feature generation.
- Developed benchmarking frameworks to evaluate different distributed computing systems and **high-performance computing** architectures, identifying optimal solutions for large-scale **ML workloads**.
- Designed and implemented an API using **FastAPI** to serve model predictions from experiments, achieving p95 latency below **200ms** for downstream applications.

Scalable Data Processing Pipeline for Research Insights

- Engineered a **Python**-based data processing pipeline on **AWS S3** and **EC2** to handle terabytes of research data, enabling efficient exploration and analysis.
- Implemented statistical analysis and hypothesis testing using **R** and **pandas** on the processed datasets, delivering key insights to research stakeholders.
- Integrated the pipeline with internal reporting tools, automating the generation of weekly summaries and reducing manual effort by approximately **15 hours** per week.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)